LEAN MODEL CANVAS

SCI-RTW: Predicting Return to Work Following Spinal Cord Injury Using Machine Learning

1. PROBLEM

- It is hard to predict whether, and when, a person with spinal cord injury (SCI) will return to work (RTW).
- Clinicians mostly rely on experience; there is no simple tool that gives an evidence-based estimate.
- Existing studies predict only yes/no RTW, not the timing, and their results are not available as a usable tool.

Existing alternatives: clinical judgement, general outcome scales, statistical models in research papers.

4. SOLUTION

- Web system with a two-layer ML model: Layer 1 predicts RTW within 10 years; Layer 2 predicts timing (Year 1, Year 5 or by Year 10).
- Each result shows a confidence level and the main factors behind it.
- Patient history, clinician notes and a printable report.
- Live online: sci-rtw-predictor.onrender.com.

8. KEY METRICS

Quality: Layer 1 accuracy 67.6%, ROC-AUC 0.744; 88% correct when confidence is high.

Usability: SUS score of 68 or more; 80% of test tasks done without help.

Adoption: active clinicians and predictions saved per month.

Key metric: predictions saved per active clinician per month.

3. UNIQUE VALUE PROPOSITION

Know if and when an SCI patient is likely to return to work, in one prediction, with the reasons behind it.

- Predicts timing, not only yes/no.
- Explainable results that support, not replace, the clinician's decision.

High-level concept: a decision-support calculator for SCI vocational rehabilitation.

9. UNFAIR ADVANTAGE

- Two-layer design trained on the large NSCISC dataset.
- Models chosen by statistical testing (McNemar), not guesswork.
- Honest evaluation: problems that inflated accuracy were found and removed.
- Tested pipeline (107 automated tests) that is easy to retrain with new data.

5. CHANNELS

Hear: conferences, physiotherapy and rehabilitation groups, UiTM Faculty of Health Sciences.

Try: hospital rehabilitation pilot through the supervisor; PERKESO Return to Work programme; free web link.

Keep using: staff training; part of the discharge and follow-up routine.

2. CUSTOMER SEGMENTS

- Rehabilitation physicians and therapists (OT, PT).
- Vocational rehabilitation / RTW case managers (e.g. PERKESO).
- SCI patients and their families (view their own results).
- Hospitals and rehabilitation centres (buyers).

Early adopters: rehabilitation doctors and RTW case managers who already plan vocational goals.

7. COST STRUCTURE

- Cloud hosting and database.
- Development, maintenance and security updates.
- Retraining and validating the model with new or local data.
- Data access and ethics approval; compliance with the Personal Data Protection Act 2010 (PDPA).

6. REVENUE STREAMS

- Annual licence per hospital or rehabilitation centre.
- Research or innovation grants for local validation.
- Free access for patients through their clinician.
- Optional paid support, training and customisation.

Numbers show the usual order for filling in a Lean Canvas (Maurya, 2012). Note: the model was trained on United States (NSCISC) data, so it must be validated on Malaysian patients before clinical use.